Iteration #03 Report

Creating a Data Visualization Dashboard for Monitoring Key Metrics and Performance Indicators in a Manufacturing Plant

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1 Dataset Description

1.1 Provide a link to the dataset you intend to use. Include a brief paragraph describing the dataset, its structure, source, and relevance to your project.

- Smart Grid Real-Time Load Monitoring Dataset

  We will mainly focus on this dataset for the project. It contains real-time load monitoring data for smart grid systems, structured as a time-series dataset with 50,000+ records recorded at 15-minute intervals. It includes key electrical parameters, renewable energy sources, environmental factors, and anomaly indicators, making it suitable for analyzing energy consumption patterns and visualization.

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Timestamp: Time of data recording.
Voltage (V), Ampere (A), Power (kW): Electrical measurements.
Reactive Power (kVAR), Power Factor: Energy efficiency indicators.
Solar & Wind Power (kW): Renewable energy input.
Grid Supply (kW): Power drawn from the grid.
Fault Indicators: Overload conditions, voltage fluctuations, transformer faults.
Environmental Factors: Temperature, humidity, and electricity price.
Predicted Load (kW): Target variable for energy forecasting.
```

  Its temporal and multivariate structure aligns well with our project goal of building a manufacturing KPI dashboard, as well as smart grid analytics, supporting research in real-time energy optimization and predictive modeling.

- Industrial Safety and Health Analytics Database

  We plan to utilize this as a secondary dataset. It contains detailed records of workplace incidents and safety compliance from various industrial environments. The database is basically records of accidents from 12 different plants in 3 different countries which every line in the data is an occurrence of an accident.

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Data: timestamp or time/date information
Countries: which country the accident occurred (anonymized)
Local: the city where the manufacturing plant is located (anonymized)
Industry sector: which sector the plant belongs to
Accident level: from I to VI, it registers how severe was the accident
Potential Accident Level: Depending on the Accident Level, the database also
                          registers how severe the accident could have been
Genre: if the person is male of female
Employee or Third Party: if the injured person is an employee or a third party
Critical Risk: some description of the risk involved in the accident
Description: Detailed description of how the accident happened.
```

  It provides structured tabular data suitable for risk-based KPI visualization, making it valuable for analyzing safety trends and operational risk factors. Integrating this dataset will allow us to extend our dashboard beyond performance metrics to include safety monitoring, helping users track and visualize incident patterns and promote safer manufacturing operations.

- If necessary, we will add some simulated data generated to mimic sensor logs and machine outputs using Python libraries like **faker, scipy.signal, statsmodels, scikit-learn**.

2 Tools and Methodologies

Our project workflow involves four key stages: data processing, statistical modeling, visualization, and deployment. The tools are selected to support efficient development at each stage:

- **Python** – Used throughout for data ingestion, transformation, modeling, and simulation.
- **Pandas, NumPy** – For tabular data wrangling and numerical operations.
- **Scikit-learn, statsmodels** – Provide time-series modeling and baseline regression analysis for load forecasting and risk modeling.
- **Plotly Dash** – Enables interactive, web-based dashboards without requiring HTML/JavaScript.
- **Tableau** – May be used to supplement static reports or generate aggregated heatmaps.
- **Jupyter / Colab** – For iterative analysis and experimentation.
- **faker, scipy.signal** – To simulate realistic sensor readings for testing visualization under dynamic loads.
- **GitHub** – Supports collaboration and version control across iterations.
- **Flask or Kafka(Optional)** – Considered if real-time streaming or service integration becomes necessary.

3 Preliminary Timeline

3.1 Draft a weekly timeline outlining the key tasks to be completed throughout the remainder of the project. Ensure the timeline includes milestones for data preparation, model development, evaluation, and final reporting.

TASK NAME	ASSIGNED TO	START DATE	END DATE	DURATION (in days)	STATUS
Team Formation and Topic Description	Qingyuan Hu, Yunmu Shu	19-May	2-Jun	14	Complete
Iteration #02	Qingyuan Hu, Yunmu Shu	2-Jun	9-Jun	7	Complete
Define metrics and find datasets	Qingyuan Hu	9-Jun	16-Jun	7	Complete
Data cleaning and preprocessing	Qingyuan Hu	16-Jun	23-Jun	7	In Progress
Provide interfaces for front/back ends	Qingyuan Hu, Yunmu Shu	23-Jun	30-Jun	7	Not Started
Develop visualizations	Yunmu Shu	30-Jun	7-Jul	7	Not Started
Develop dashboard wireframes	Yunmu Shu	7-Jul	14-Jul	7	Not Started
Final dashboard prototype with interactivity	Yunmu Shu	14-Jul	21-Jul	7	Not Started
Debug and evaluation	Qingyuan Hu, Yunmu Shu	21-Jul	28-Jul	7	Not Started
Final submission with report and documentation	Qingyuan Hu, Yunmu Shu	28-Jul	4-Aug	7	Not Started
Prepare for presentation	Qingyuan Hu, Yunmu Shu	4-Aug	11-Aug	7	Not Started